

EXPLORATORY DATA ANALYSIS OF SAAS PRODUCT USAGE & CUSTOMER RETENTION

Subscription Behaviour, Product
Usage & Customer Retention



TOOLS USED

Python, Pandas, Matplotlib, Seaborn & Plotly

PRESENTED BY

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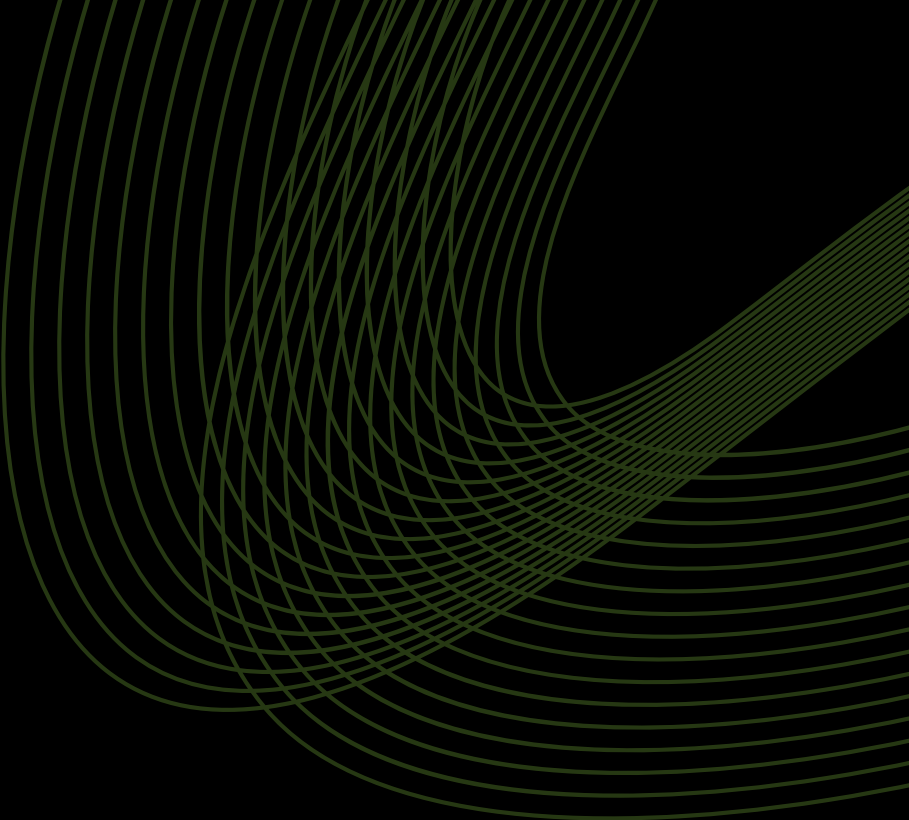


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PROBLEM STATEMENT & OBJECTIVE

Problem Statement:

Modern SaaS businesses operate across diverse customer segments with varying usage and retention behaviors, making it challenging to pinpoint the engagement and operational factors that influence churn and long-term revenue.

Objective:

To explore the SaaS customer usage and retention dataset to uncover patterns, insights, and trends that can help businesses make data-driven decisions around customer engagement, retention, and revenue growth.



EDA WORKFLOW

For this analysis, a structured workflow was followed, involving data collection, understanding, cleaning, exploration, and summarization of insights to allow a clear understanding of the dataset and its trends.

01

Data Collection

- Gathered the Google Play Store dataset from Kaggle

02

Data Understanding & Anomaly Detection

- Looked at data distributions
- Found missing values, outliers, and unusual patterns

03

Data Cleaning & Treatment

- Fixed missing or incorrect values
- Standardized formats for consistency

04

Exploratory Analysis

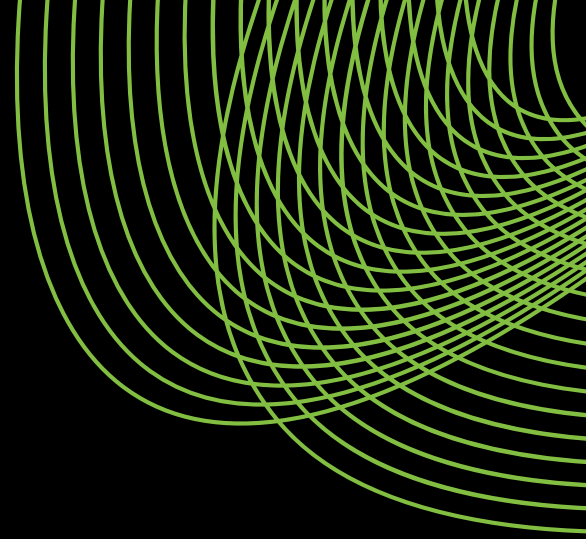
- Univariate Analysis: e.g Customer Segment, Complaint Type, Contract Type etc
- Bivariate Monthly Logins vs Features Used
- Multivariate Analysis: CSAT vs Contract Type vs Customer Segment

05

Insights & Reporting

- Summarized patterns and trends
- Highlighted key findings for developers and businesses

KEY QUESTIONS EXPLORED



- How do user demographics (age and gender) influence shopping behavior and engagement patterns on the platform?
- Which stages of the shopping journey (browsing, cart, checkout) contribute most to drop-offs and missed conversions?
- Do occasional shoppers present a greater growth opportunity compared to already frequent buyers?
- How effective are product recommendations in influencing purchase decisions, and where do they fall short?
- What role does pricing perception play in cart abandonment and cross-platform comparison behavior?
- How do customer support responsiveness and delivery reliability impact overall satisfaction and trust?
- How do reviews and post-purchase experiences influence future purchase intent and platform credibility?
- Which satisfaction drivers most strongly contribute to repeat purchases and long-term loyalty?



DATA OVERVIEW

The dataset provides insights into SaaS customer behavior, covering demographics, subscription characteristics, product usage, billing activity & support interactions. It also captures customer engagement signals, satisfaction metrics, and feedback indicators that are relevant for understanding retention and churn dynamics.

Data Source: Kaggle

Dataset Size

10000

Records

32

Features

Purchase Diversity

3

Customer Segments

7

Countries

3

Contract Types

